Shisham Adhikari

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Citizenship: Nepal *Visa:* F-1 *Address:* Davis, CA, USA

EDUCATION

University of California, Davis (UC Davis) — Davis, CA
Ph.D. in Economics — May 2026 (expected)
M.S. in Economics *(Best First-Year Student Award)* — September 2022

Reed College — Portland, OR
B.A. in Mathematics and Economics *(Phi Beta Kappa Inductee)* — May 2021
Thesis: A Ramsey Model with Heterogeneous Rates of Return: How do investment return differentials affect inequality?

Carnegie Mellon University — Pittsburgh, PA
Public Policy and International Affairs (PPIA) Fellow *(Data Analytics Track)* — August 2019

RESEARCH FIELDS

Primary: Macroeconomics, Labor Economics, Climate Economics
Secondary: Search–Match Theory, Time-Series Econometrics

REFERENCES

Prof. Athanasios Geromichalos
University of California, Davis
Associate Professor of Economics
ageromich@ucdavis.edu

Prof. Katheryn Russ
University of California, Davis
Professor of Economics
keruss@ucdavis.edu

Prof. Oscar Jordà
University of California, Davis
Professor of Economics
ojorda@ucdavis.edu

PUBLICATIONS

Adhikari, Shisham, Geromichalos, Athanasis, Gürsoy, Ateş, and Kospentaris, Ioannis, "How Much Work Experience Do You Need to Get Your First Job?", (2025). *Review of Economic Dynamics*, 58, Article 101301.
Paper | Appendix | Slides

WORKING PAPERS

Adhikari, Shisham, "Resolving Coordination Frictions in Structural Labor Transitions: The Case of the Green Transition." (2025) Job Market Paper.
Presented at: IMF Climate Analytics Hour; EER Conference on Macroeconomic and Financial Aspects of Climate Change (Mexico City, 2025); CAFIN 2024 (UC Santa Cruz); UC Davis Macro/International Seminar.

Adhikari, Shisham, "Task-Level Policy for the Green Transition: Moving Beyond Binary Classifications." (2025).

Adhikari, Shisham, Batibeniz, Fulden, Bellon, Matthieu, Massetti, Emanuele, Seneviratne, Sonia, and Waidelich, Paul, "Projecting Macroeconomic Cost of Climate Change: A Case for Adaptation." (2025).

Adhikari, Shisham, and Guo, Si, "Search Frictions and Industrial Share." (2025).

WORK EXPERIENCES

International Monetary Fund (IMF) — Washington, DC
PhD Fund Intern – FADCP — June 2025 - September 2025

- Studied adaptation assumptions in climate–economy losses and built projection framework for future impacts.
- Developed global grid-level geo-spatial datasets to quantify macroeconomic costs of extreme weather events.

RStudio PBC — Remote
Data Science Educator — September 2021 - December 2022

- Built interactive R tutorials and projects for RStudio Academy to train professionals in data science.

Software Developer (Intern) — May 2021 - August 2021

- Developed an R package (shinymodels) for interactive model exploration to facilitate model deployment.

TECHNICAL SKILLS

Programming: R, Python, MATLAB, Stata, QGIS, SQL
Tools & Platforms: Jupyter, RStudio, Git/GitHub, UNIX/Linux, Microsoft Office, LaTeX, GIS tools
Languages: Nepali (native), English (fluent), Hindi (fluent), Spanish (intermediate)

RESEARCH EXPERIENCES

University of California, Davis Davis, CA
Graduate Student Researcher for Prof. James Bushnell and Prof. Aaron Smith June 2024 – March 2025
Graduate Student Researcher for Prof. Kathryn Russ June 2023 – September 2023

Reed College Portland, OR
Undergraduate Thesis, Advisor: Prof. Jasmine Jiang August 2020 – May 2021
Research Assistant, Advisor Prof. Tristan Nighswander May – August 2020
Research Assistant, Advisor Prof. Kimberly Clausing June – August 2019

CFA Institute Research Challenge Portland, OR
Team Leader, Advisor: CFA JB Groh October 2018 – March 2019

TEACHING EXPERIENCES

University of California, Davis Davis, CA
Graduate Teaching Assistant: Economic and Financial Forecasting, Econometrics, Game Theory, Intermediate Macroeconomics, Intermediate Microeconomics, Principles of Microeconomics, Principles of Macroeconomics

Reed College Portland, OR
Undergraduate Teaching Assistant: International Monetary System, International Trade, Behavioral Finance, Econometrics, Data Science, Probability, Math Statistics, Calculus, Analysis, Linear Algebra, Vector Calculus

LEADERSHIPS AND INVOLVEMENTS

California Council on Science and Technology (CCST) Sacramento, CA
Translator, 2025 CCST Science Translators Showcase January 2025 – March 2025

- Presented at the California State Capitol to legislative staffers during CCST's Science & Technology Week.
- Completed science communication training focused on translating technical research into policy-relevant insights.

Girls Who Invest Scholar New York, NY
Scholar/Alumni Ongoing

- Accepted into a selective program focused on understanding the finance industry and core investment concepts.

PERIOD. Remote
Data and International Policy Analysis Intern May 2020 - August 2020

- Applied machine learning to classify global menstrual health policies and support policy recommendations.

Reed College Finance and Investment Club Portland, OR
President December 2018 - December 2019

- Managed a $75K investment portfolio, achieving a 15% increase during tenure.

SCHOLARSHIPS, GRANTS, HONORS AND ACHIEVEMENTS

Department Research/Travel Award ($8,000)	UC Davis 2023
Best First-Year Student Award ($500)	UC Davis 2021
Carl Burgdorfer CFA Scholarship ($2,000)	CFA Society of Portland 2021
Evan Rose Research Fund ($6,500)	2021
Gerald M. Meier Award for Distinction in Economics ($500)	2021
Public Policy and International Affairs (PPIA) Fellowship ($3,000)	Carnegie Mellon University 2020
Summer Internship Award ($4,000)	2020
Collaborative Research Grant ($2,488)	2020
Economics Summer Research Award ($4,900)	Reed College 2019
Career Advancement Fund ($1,265)	2019-2020
Student Opportunity Subsidy (SOS) Grant ($700)	2019
Davis Projects for Peace Award ($10,000)	2018

CERTIFICATIONS AND AFFILIATIONS

Climate Change and AI (CCAI) Summer School	Montreal, 2024
Journal of International Economics (JIE) Summer School	Crete, 2023
Commendation for Academic Achievement	Reed College, 2017-2021
Foundations of Investment Management	Wharton Online, Jan 2020
Accounting, Analyzing Financial Reports, and Excel Training	Wall Street Prep, Oct 2019
The Investment Foundations Program	CFA Institute, Oct 2019

ABSTRACTS

Adhikari, Shisham, "Resolving Coordination Frictions in Structural Labor Transitions: The Case of the Green Transition", (2025); *Job Market Paper*; Draft | Slides

Ambitious decarbonization requires not only technology and capital, but also rapid reallocation of workers into green jobs. This reallocation faces a two-sided coordination failure: workers hesitate to pay entry costs without job prospects, while firms underinvest in vacancies without a ready talent pool. This friction compounds the environmental externality and stalls the transition when both speed and scale are essential. I develop a two-sector Diamond–Mortensen–Pissarides model with costly worker entry, firm vacancy creation, and balanced-budget instruments to compare worker-side support, firm-side support, and coordinated dual policy. Calibrated to a U.S. energy transition from 2% to 14% green employment by 2030 (24 million jobs), I find three results. First, coordinated policy dominates one-sided alternatives, reaching the target with 39 basis points lower unemployment, 5 basis points of GDP lower fiscal cost, and 23 basis points higher welfare. Second, optimal policy is dynamic: total support is hump-shaped, and the instrument mix rotates worker–firm–worker as the binding constraint shifts along the transition path. Third, coordinated policy raises aggregate welfare once the environmental benefit exceeds a consumption-equivalent gain of 0.87%. The main lesson is that climate policy design, not just scale, matters for socially inclusive and fiscally sustainable transitions.

Adhikari, Shisham, "Task-Level Policy for the Green Transition: Moving Beyond Binary Classifications", (2025); Draft

This paper develops a task-based general-equilibrium model to evaluate targeted green-input subsidies, offering new insights into fiscal policy for the green transition. Production is represented as a continuum of tasks ranked by "greenness," moving beyond the conventional "green vs. dirty" sectoral split. By capturing task-level heterogeneity, the model shows that competitive markets allocate too few tasks to green inputs relative to a planner who internalizes environmental externalities, justifying corrective subsidies. The framework addresses three policy-relevant questions: (i) *Design*—how should subsidies vary across tasks to maximize reallocation toward green methods? (ii) *Trade-offs*—what productivity cost, if any, accompanies environmental gains? (iii) *Financing*—which tax base funds subsidies with the least distortion? Key findings are: (1) subsidies work best when the productivity gap between green and traditional inputs is small, (2) in a U.S. calibration, welfare improves only if the externality parameter is roughly three times the standard public-goods benchmark (1.2 vs. 0.4), implying a 4.3 % consumption-equivalent threshold, and (3) lump-sum taxes impose the smallest welfare loss, followed by capital taxes, while labor-income taxes are most distortionary.

Adhikari, Shisham, Batibeniz, Fulden, Bellon, Matthieu, Massetti, Emanuele, Seneviratne, Sonia, and Waidelich, Paul, "Projecting Macroeconomic Cost of Climate Change: A Case for Adaptation." (2025).

A large empirical literature has linked weather shocks to GDP to estimate future losses from climate change and has obtained a wide range of results. We develop a framework that nests past approaches, distinguishing between modelling assumptions about how changes in climate variables translate into GDP impacts. We show how four prominent model choices in the literature are related to different assumptions about the persistence of impacts and adaptation. We follow recent developments and apply our framework to extreme weather shocks, including extreme heat and harsh droughts, using high-resolution climate projections to obtain long-term GDP impacts. We compare results under the four different modelling assumptions and show that they play an outsized role in determining the size of impacts. Finally, we quantify the role of different sources of uncertainty.

Adhikari, Shisham, and Guo, Si, "Search Frictions and Industrial Share." (2025).

We study industrial development in a two-sector Diamond–Mortensen–Pissarides model with capital-intensive manufacturing. Since firms commit capital before wage bargaining, a holdup problem depresses investment, while search externalities keep manufacturing employment inefficiently low. The social planner invests more in capital and reallocates labor toward manufacturing; quantitatively, capital deepening raises output but can reduce employment relative to the decentralized outcome. We show that a zero–net-cost mix of capital tax credits, training subsidies, and vacancy license fees replicates the planner's allocation, highlighting the need for coordinated support to both firms and workers.